Multi-User Behavioral Privacy Filtering for mmWave Radar Sensing

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***Abstract*—As an advanced technology for non-contact sensing, mmWave radar enables fine-grained measurement of a wide variety of user behaviors. While creating intelligence and convenience, it also concerns behavioral privacy and security, as radar signals contain a wealth of behavioral information. Existing solutions are either incapable of customizable privacy protections or cannot cope with multi-person scenarios. This paper presents a Multi-user behavioral privacy Filter, MuFilter, a data masking system centered on the idea of dimensional signal interference. It determines the sensing signatures that need to be preserved or interfered with based on the sensing services that users want to enable and disable, thereby making targeted tampering on the radar signal. On this basis, we introduce the multi-person tracking technology to allow MuFilter to determine the number of users in unknown scenarios. Moreover, a subspace tampering technique is proposed to ensure that each tampering only affects the target user and not other users, thus supporting personalized privacy protection for multiple users. Experiments show that MuFilter can interfere with targeted behavioral signatures with a 100% success rate, while the degree of impact on other users' signatures ranges from 0% to 3.85%.**

***Index Terms*—mmWave radar, sensing security, behavioral privacy, data masking.**

I. Introduction

A. Motivation & Problem Statement

NON-CONTACT human sensing technology collects and monitors human-related data conveniently without direct contact with the human body. It is applicable to smart homes, healthcare, and security surveillance, offering users a more comfortable experience and opening up possibilities for smarter living. In recent years, Millimeter-Wave (mmWave) sensing technology has supported a wide range of human sensing applications due to its excellent sensing performance and low deployment costs [1]. However, the rapid development of mmWave technology, while bringing convenience to people's lives, has also prompted concerns about privacy and security [2]. In this paper, we focus on privacy issues of user behavior resulting from the leakage of radar raw data. The radar data supports a variety of services such as localization & tracking [3], posture estimation [4], [5], hand & gesture sensing [6], [7], activity recognition [8], vital signs monitoring [9], etc. Whether or not the user needs and turns them on, these sensing services can be realized by data processing of the raw radar signals, which raises behavioral privacy issues as the user may not want to expose all the information on their behavior. For example, a user may want radar to provide in-air writing and respiration monitoring services, but does not want to give away their position and posture information. Further, with multiple users in the sensing area, their respective privacy needs are more diverse or even conflicting. Therefore, this article aims to *achieve customizable behavioral privacy protection for multiple users, allowing each user to freely set the enabled sensing services and hidden behavioral information*.

B. Limitations of Prior Art

Some studies propose to deceive external radar detection from the physical layer. They place specific tags [10] or backscatter arrays [2], [11] in the environment to modulate the radar reflection signals and thus spoof radar measurements of range, angle, and velocity. This approach *only interferes with the detection of human presence and cannot achieve the retention and hiding of specific behavioral information*. mmFilter [12] proposes the idea of dimensional signal interference, which implements various data tampering methods based on the signature dimensions that different sensing applications rely on, achieving customizable behavioral privacy protection. However, it only considers single-user scenarios, and two reasons constrain mmFilter from being deployed in multi-user scenarios. On the one hand, it *lacks the capability to detect and separate multiple users in the scene*. On the other hand, mmFilter's data tampering methods act on the global, meaning that tampering for one user will inevitably affect other users, and thus it *cannot support personalized privacy protection settings for multiple users*.

C. Our Approach

To overcome existing limitations, we propose MuFilter, a multi-user behavioral privacy filter for mmWave sensing technology. Its core idea follows the dimensional signal interference. Utilizing the invertibility of Fast Fourier Transform (FFT), we spatially unfold the radar data, tamper with the information in some dimensions, and then transform the tampered data into

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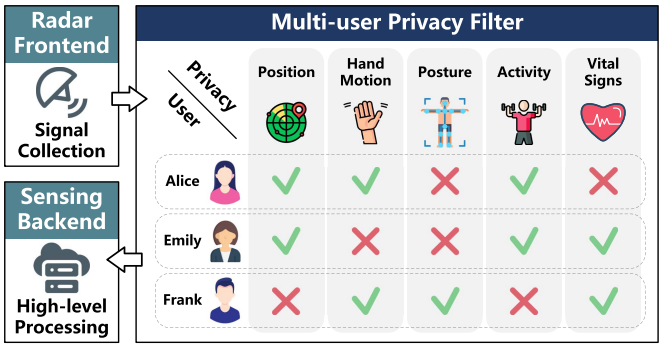


Fig. 1. Illustration of the multi-user behavioral privacy filter.

the raw signal form by Inverse FFT (IFFT). In this way, we avoid behavioral information leakage, as it has already been tampered with at the data source of the sensing services. In order to serve multiple users, we introduce mmWave multi-person tracking technology to create spatial records for each user. Then, we utilize the spatial segmentation capability of mmWave radar to isolate each user and perform signal tampering in subspaces, thereby limiting the area of impact to the vicinity of the user's location. As shown in Fig. 1, MuFilter can be deployed in software or hardware between the Analog-to-Digital Converter (ADC) and sensing processors. It can provide users with customizable privacy protection for five types of behavioral information: position, hand motion, posture, activity, and vital signs.

D. Novelty & Advantages

MuFilter is the first system to implement multi-user customizable behavioral privacy filtering, and it has two main advantages over previous works. Compared with physical-layer deception methods, MuFilter achieves targeted privacy protection, with various data tampering methods being compatible with each other. Compared with another data layer signal tampering method mmFilter, MuFilter can *effectively deal with multi-user scenarios, achieving personalized information hiding for multi-user without affecting each other*. In other words, the main novelty of MuFilter is that it solves two challenges: *how to determine the number of users in unknown scenarios*, and *how to independently interfere with the behavior information of each user*.

This achievement is largely attributed to two capabilities of our system. (1) Multi-person tracking. In mmWave multi-person sensing, it is essential to determine the location of each user and to isolate their signals. MuFilter adopts the point target tracking framework based on spatial point clouds. The Range-Azimuth (RA) point cloud obtained from Multiple-Input-Multiple-Output (MIMO) array signals is clustered to detect human targets. User trajectories and new clusters will be correlated and updated between frames using the Hungarian algorithm and Kalman filtering. We also use a state mechanism for trajectory management. The obtained user positions will contribute to multi-user signal isolation and targeted tampering. (2) Subspace signal tampering. Tampering based only on data dimensions affects all targets globally, which goes against the need for multi-user personalization. To solve this, MuFilter fully utilizes the spatial separation capability of mmWave radar. We use Range-Angle FFT to slice the space into RA bins, and then extract the subspaces around each user based on the tracking results. Signal tampering will be performed on one subspace after another, and the tampered subspaces will then be restored to their original positions and eventually uniformly inverted to the original signal format by full-space Angle-Range IFFT. We also solve collision problems that can occur in subspace separations. In this way, data tampering for one user is prevented from affecting other users. Experiments show that MuFilter can interfere with targeted behavioral signatures with a 100% success rate, while its impact on the signatures of other users ranges from 0% to 3.85%.

The remainder of this paper is organized as follows. Section II provides the principle of radar-based human sensing and the threat model. Section III presents the system overview of MuFilter. Section IV demonstrates the customizable behavioral privacy filter in single-user scenarios, and Section V describes the methodology to achieve multi-user personalization. We evaluate the performance of MuFilter in Section VI. The related works are summarized in Section VII, and Section VIII concludes this paper.

II. Preliminaries

This section introduces the preliminaries of this work, including the working principle of mmWave radar, typical sensing applications and corresponding dependency signatures, and the threat model.

A. mmWave Working Principles

mmWave radar detects the distance, velocity, and angle of objects by transmitting the frequency-modulated continuous wave (FMCW) signal known as the chirp signal, with the frequency increasing linearly over time. Suppose the chirp signal starts at the initial frequency f_c, linearly increases over time T_c, and ends at $f_c + B$ where B is the bandwidth, as illustrated in Fig. 2. The frequency of the chirp signal can be expressed as $f(t) = f_c + kt$, where the slope k is given by $k = \frac{B}{T_c}$. The transmitted (Tx) signal can be expressed as

$$S_T(t) = A_T e^{j2\pi\left(f_c t + \frac{B}{2T_c}t^2 + \phi_0\right)}, \tag{1}$$

where A_T is the amplitude of Tx signal, and ϕ_0 is the initial phase, which is a constant. The Tx signal will be reflected by objects and back to the radar. The received (Rx) signal is a delayed version of the Tx signal, with a time delay given by $\tau = \frac{2d}{c}$, where c is the speed of light and d is the radar-object distance. The Rx signal can be expressed as

$$S_R(t) = A_R e^{j2\pi\left(f_c(t-\tau) + \frac{B}{2T_c}(t-\tau)^2 + \phi_0\right)}, \tag{2}$$

where A_R is the Rx signal amplitude after attenuation. Then, the Tx and Rx signal are mixed to obtain the Intermediate Frequency (IF) signal

$$S(t) = A_{IF} e^{j2\pi\left(f_c\tau + \frac{B\tau}{T_c}t - \frac{B\tau^2}{2T_c}\right)}. \tag{3}$$

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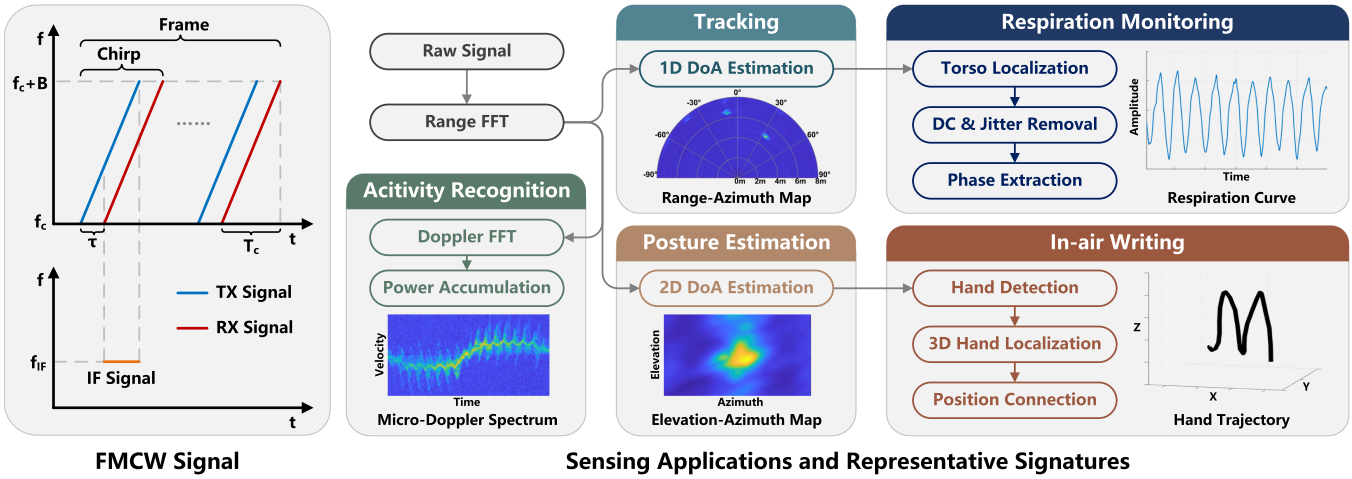


Fig. 2. Principles of mmWave radar human sensing.

The time interval consisting of multiple chirps is called a frame, which is the basic unit of radar signal processing.

B. Sensing Applications and Signatures

This paper involves five commonly used human sensing applications. While introducing them, we focus on the sensing signatures they rely on and the ways in which these signatures are obtained, which are depicted in Fig. 2. The sensing signatures contain essential information for their corresponding applications, and their illustrations will intuitively demonstrate the interference effects of the MuFilter system.

Tracking: Tracking is a crucial application and serves as the foundation for other sensing applications. The resulting trajectory from tracking contains the user's position at any moment, including two signature dimensions: range and azimuth. Therefore, its sensing signature can be represented by the RA Map (RAM). For the range dimension, an FFT is usually conducted to the IF signal to separate reflective objects, which is known as Range FFT. For the angle dimension, direction of arrival (DoA) estimation is applied to obtain the azimuth of the user. The most commonly used DoA estimation method is to perform FFT on the antenna array signal, which is called Angle FFT. Another method is Capon beamforming which minimizes the variance distortion response to enhance the desired signal while minimizing interference and noise. Capon beamforming has higher resolution and stronger anti-noise capabilities. Therefore, although our MuFilter system can act on any DoA estimation algorithm, for the sake of writing simplicity and demonstration purposes, the sensing applications in this paper will solely utilize Capon beamforming and showcase its results.

Posture Estimation: The user's posture estimation in 3D space, including skeleton [4] and mesh [5], can vividly depict the human body shape. The posture involves horizontal and vertical information in 3D space, therefore the Elevation-Azimuth Map (EAM) can be used as the sensing signature for posture estimation. 2D DoA estimation is needed to obtain information in the two angular dimensions. Similar to 1D DoA estimation, 2D Angle FFT and 2D Capon beamforming are the most commonly used 2D DoA estimation methods.

In-air Writing: As one of the typical applications of human-computer interaction, in-air writing tracks the user's hand movement and converts it into text or graphic input, which can be applied to scenarios such as virtual reality and in-air whiteboards. Regardless of the specific method of in-air writing, it must detect the position of the hand in 3D space. Both in-air writing and posture estimation require azimuth and elevation information. The difference lies in the observation targets: in-air writing focuses solely on the hand, while posture estimation encompasses the entire body. Therefore, based on practical needs, we make the following reasonable preconditions: a user who is unwilling to expose their hand trajectory is also unwilling to reveal their posture information. Conversely, if a user is willing to provide their posture information, they would also be willing to perform in-air writing detection.

Activity Recognition: The user's behavior and intentions can be inferred through recognizing and analyzing user's motion activities. Activity recognition is usually accomplished by training neural networks using micro-Doppler spectrum [13]. The phase change of the chirp signal represents the displacement of the reflector and can be used to measure instantaneous velocity. By conducting FFT on signals from multiple chirps within a single frame, which is known as Doppler FFT, the velocity spectrum for this frame is obtained. After performing Doppler FFT across multiple frames and stitching the velocity spectrum together, the micro-Doppler spectrum is generated, which contains rich velocity information of the user's motion.

Respiration Monitoring: We can gain insights into the user's health status and perform exercise management through respiration monitoring, which plays a significant role in elder healthcare and fitness domains. When a person breathes, there is a tiny periodic displacement in the chest and abdomen which can be measured by analyzing the phase change of the signal. Therefore, the respiration curve can be regarded as the sensing signature. The first step to obtain the respiration curve is to locate the torso containing the chest or abdomen. Next, mitigate the effects of direct current component (DC) and jitter. Finally, the phase is extracted and converted into the respiration curve.

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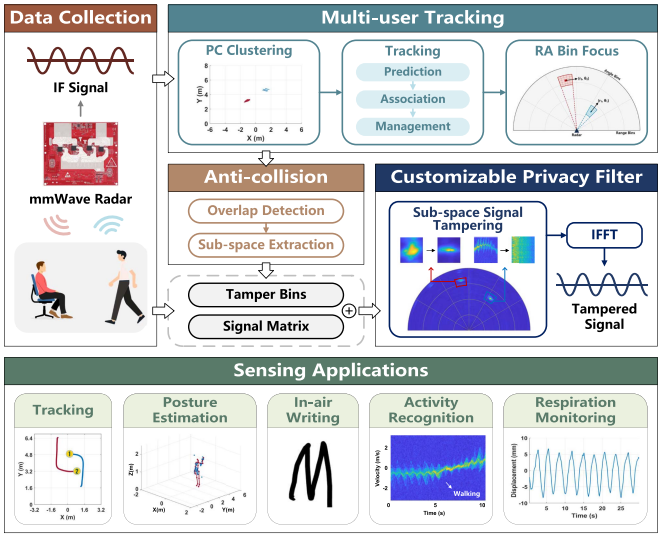


Fig. 3. System overview of MuFilter.

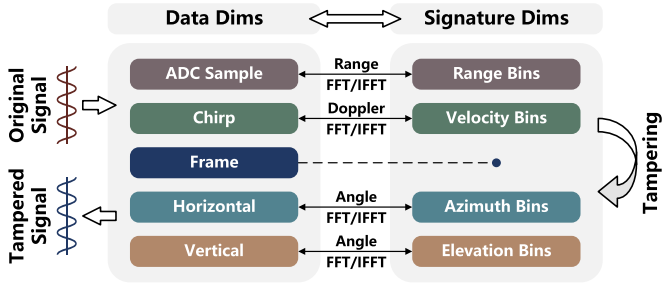


Fig. 4. Principle of dimensional data interference.

C. Threat Model

Attack Capability. Cyberattack Capabilities: Attackers can gain control of the sensing backend and obtain all radar data streaming into the backend. Attackers cannot bypass the privacy filter to obtain the original radar data. Attackers cannot modify the privacy filter's settings and influence its actions.

Sensing Capabilities: Attackers know all the radar parameters, including chirp slope, ADC sampling numbers, frame period, antenna layout, etc. Attackers know the radar data format and can convert the data stream or files into the signal matrix. Attackers can implement each sensing application in multiple ways.

2) Attack Goal: Attackers want to obtain and monitor users' behavioral information on position, posture, hand motion, activity, and vital signs.

3) Defense Guideline: The defense should be effective for most sensing methods of a certain kind of behavioral information. The data perturbation should be restrictive and not affect the acquisition of allowed behavioral information. The computational burden should be low to reduce deployment costs and maintain real-time data.

III. System Overview

To achieve personalized privacy protection in multi-user scenarios, we propose the MuFilter system, the overview of which is shown in Fig. 3. The mmWave radar collects users' reflected signals and outputs digital IF signals, which will be passed to the hardware or software system equipped with MuFilter. MuFilter first obtains RA point clouds and performs multi-target detection through Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering. Then, we get each user's trajectory through a multi-target tracking process involving Kalman filtering and the Hungarian algorithm. Next, MuFilter performs Range FFT and Angle FFT on the raw radar data to obtain a reversible RA signal matrix. We map the user position obtained by tracking into the RA signal matrix, and expand the users' RA bins to cover the entire body. The expanded bins are then input to the collision processing module to detect the spatial overlap of users and determine the appropriate tamper bins based on the anti-collision mechanism. Then, we extract the sub-signal matrix contained in each user's tamper bins and perform personalized tampering according to the user's privacy protection needs. Finally, the tampered signal matrix will be converted to raw radar signal format through Angle and Range IFFT. Our system is capable of multi-user customizable behavioral privacy protection across five sensing applications: tracking, activity recognition, posture estimation, in-air writing, and respiration monitoring.

IV. Customizable Behavioral Privacy Filter

This section demonstrates customizable behavioral privacy filter for five sensing applications in the single-user scenario. We begin with an elaboration of the core concepts of dimensional data interference and signal reversion, and then introduce and review data tampering methods for each signature dimension.

A. Core Concept

In general, different sensing applications rely on different signature dimensions, which makes it possible to implement customizable privacy protection. The basic idea is to interfere with the signature dimensions on which the disabled application depends, so as to protect specific behavioral information without affecting other sensing applications, which is called dimensional data interference. The implementation of this idea is based on another important concept, signal reversion, which utilizes the invertibility of FFT. That is, data can be converted from "time domain" to "frequency domain" by FFT, or from "frequency domain" back to "time domain" by IFFT.

As shown in Fig. 4, the radar data has five data dimensions, three of which are temporal (ADC sample, chirp, and frame) and two of which are spatial (horizontal and vertical antennas). After FFT, these data dimensions are transformed into range, angle, short-time, and long-time signature dimensions for subsequent sensing applications. We are then able to eliminate or change the corresponding behavioral information contained in the radar data by performing data tampering in specific signature dimensions. Finally, with IFFT, the radar data is converted from the signature dimensions back to the data dimensions, its original data format. In this way, we achieve targeted data masking while

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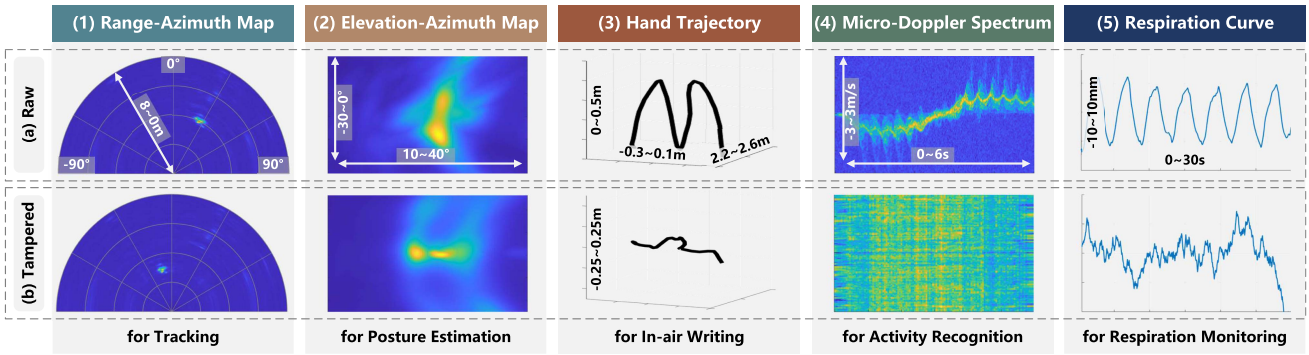


Fig. 5. Demonstration of behavioral privacy filtering in single-user scenarios.

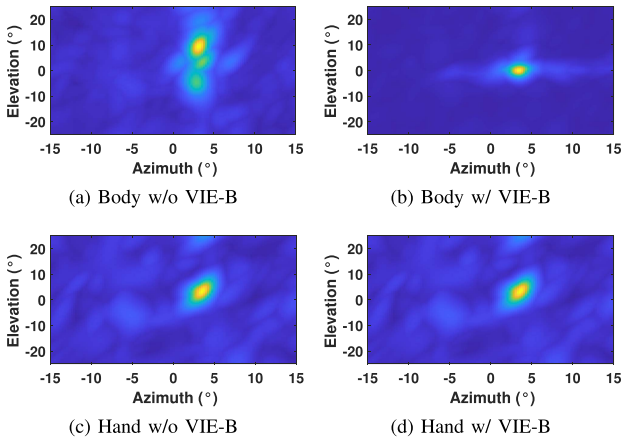
maintaining the data integrity. This approach shares a similar principle with differential privacy [14], [15] by protecting user privacy by adding noise to the data, where the amount of noise is crucial as it must effectively protect user privacy without causing the data to become unusable due to excessive noise.

B. Tampering Method

A total of five dimensional data interference methods or modes have been proposed for five typical applications: tracking, posture estimation, in-air writing, activity recognition, and respiration monitoring. Rows (a) and (b) of Fig. 5 respectively show the sensing signatures before and after tampering.

Position Reset: Position information mainly contains two dimensions, range and azimuth. In the hiding of the position, the integrity and continuity of the user's signal need to be maintained, as the position information is the basis for other sensing applications and guides the subsequent signal extraction. Therefore, the Position Reset (PR) is proposed. It first performs Range FFT and then 1D Angle FFT with the number of FFT points as the number of horizontal array elements. After determining the user's position in the RA domain, the RA bins near the user will be moved to a preset position. For example, in Fig. 5(Column 1), the user's position is reset to (3 m, $-20°$). Then, the Angle IFFT and Range IFFT are performed to convert the azimuth bins and range bins to horizontal array signals and ADC sampled signals. In this way, the user's real position is eliminated, but the signal near the fake position retains the user's full local temporal and spatial information, thus other functions such as activity recognition and respiration monitoring can be conducted normally.

Vertical Information Erasure: Posture estimation and in-air writing are obviously inseparable from elevation estimation, or in other words, the vertical information. Thus, when we want to protect information of posture or hand motion, we can perform Vertical Information Erasure (VIE). Specifically, we replicate signals of a horizontal array in the 2D array as signals of horizontal arrays at other heights, so that the horizontal signals at each height are the same, which will completely invalidate the vertical dimension. As shown in Fig. 5(Columns 2∼3), VIE compresses the spatial information onto the plane of 0° elevation. In the elevation-azimuth map, the highlighted area



(a) Body w/o VIE-B (b) Body w/ VIE-B

(c) Hand w/o VIE-B (d) Hand w/ VIE-B

Fig. 6. The impact of VIE-B on an in-air writing user's EAM.

undergoes deformation: while the azimuth information which indicates user's position is retained, the height information of the body is removed. In the hand trajectory, variations in hand height are eliminated, resulting in a distorted trajectory that lies on a horizontal plane, which differs significantly from the actual hand movement. Note that the posture information includes hand motion, thus when a user wants to hide their hand motions, they are likely to want to hide their posture. With this in mind, we consider two scenarios. (1) The user wants to hide posture and hand motion. The signal replication can be performed directly to eliminate all vertical information from the signal, which we call **VIE-A** mode. (2) The user wants to hide posture but retain hand motion. In this case, we need to perform Range FFT to separate the hand and body in the range dimension, and then skip the range bins of the hand when performing signal replication, which we call **VIE-B** mode. Fig. 6 shows the impact of VIE-B on vertical information. The sensing object is an in-air writing user, whose EAMs of the body (with the extended hand separated for demonstration) and the extended hand are shown as Fig. 6(a) and (c), respectively. After VIE-B, the EAMs change to Fig. 6(b) and (d), which shows that the body is erased and the hand is preserved.

Chirp Disorganization: The Doppler FFT converts the phase changes of chirp signals into a velocity spectrum, the crucial information for activity recognition. It is easy to interfere with

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this information by simply changing the chirp ordering in each frame, so that the phase changes cannot align with the time changes. We refer to this method as Chirp Disorganization (CD). As shown in Fig. 5(Column 4), after chirp disorganization, the velocity information is completely erased. This is because the correlation between the chirps is broken, feeding the wrong sampling order into the FFT. The temporal disturbance of the data in the time domain caused an energy disturbance in the frequency domain, preventing the user's instantaneous velocity information from being captured. In addition to this method, fine-grained data modifications can be performed on velocity bins and then convert velocity bins back to chirp signals by Doppler IFFT.

Inter-frame Phase Fluctuation: Since frames are the basic unit of radar signal processing, they can be considered as a long-time dimension independent of the chirp dimension. Frames carry the objective temporal logic. We cannot simply reorder it like we do with chirps, as this will cause confusion in all sensing applications. Fortunately, for complex signals in one frame, superimposing a phase on them does not affect the signal processing result, i.e., $F(e^{j\phi}S) = F(S)$, where F represents the signal processing operations and ϕ is the superimposed phase. When a random phase is superimposed on each frame, the inter-frame phase correlation is erased without affecting the intra-frame signal processing, which is called Inter-Frame Phase Fluctuation (IFPF). This method can interfere with applications that rely on inter-frame phase coupling such as respiration monitoring. As shown in Fig. 5(Column 5), the disruption of phase continuity simultaneously disrupts the continuity of the respiration curve, therefore the sensing result is a meaningless noise curve.

The above methods have been proven to be independent and compatible [12]. Once the attacker acquires the tampered raw signal and performs FFT on it, they will obtain results that have been manipulated by MuFilter. The attacker is unaware of the composition, sequence, and parameters of the tampering methods, and lacks access to process variables involved during tampering. Consequently, the attacker cannot recover the original information, even if they recognize that the data has been altered. Brute-force cracking is also difficult to restore data for the following reasons. (1) The correctness of the cracking attempt cannot be verified. The proposed privacy filter is not an encryption system with a clear password, but an entropy-increasing process for the information contained in radar data. Each cracking attempt yields a seemingly correct answer, but it is uncertain which answer is true. For example, the PR moves the user's RA bins from the real position to a fake position. Reverse attempts to the true position always yield reasonable results because RA bins can be moved from any position to the false position. Similarly, based on the phase noise sequence caused by the IFPF, an arbitrary respiration curve can be generated by resetting the phase, but it is uncertain which one is the user's respiration curve. Therefore, the inherent nature of the privacy filter makes brute-force cracking lose the critical attempt feedback. (2) The probability of a single attempt being correct is extremely low. Attackers obtain as many sensing results as cracking attempts. Therefore, due to the lack of feedback on cracking attempts, attackers have only one chance to guess the privacy filter's settings, and the probability of this guess being correct is very low. For example, the probability of correctly guessing the real order of 128 chirps is as small as $1/128! \approx 2.6 \times 10^{-216}$. Note that this is only the probability of cracking the single CD technique. Considering the combinations of multiple tampering techniques, the probability of restoring the entire privacy filtering process will be even much smaller than 2.6×10^{-216}. Thus, attackers are almost unable to correctly restore the tampered signal in a single attempt.

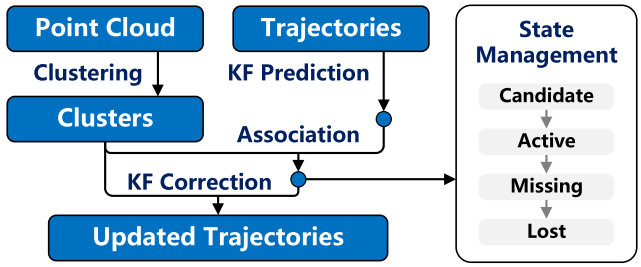


Fig. 7. Multi-person tracking workflow.

V. Achieving Multi-User Personalization

In this section, we extend the customizable behavioral privacy filter to multi-user scenarios. Undoubtedly, the greatest difficulty stems from the spreading effect of signal interference, thus requiring us to effectively isolate each user's signal. Our solution utilizes the spatial separation capability of radar. MuFilter first performs multi-person tracking to obtain user positions. Then, extract spatial bins near each user to provide further spatial boundaries for dimensional data interference. Ultimately, the data tampering is performed only within the subspaces, thus preventing cross-interference between users.

A. Multi-Person Tracking

Benefiting from the development of mmWave MIMO technology, mmWave radars nowadays have excellent enough angular resolution. From standard steps of Range FFT, 1D DoA estimation, and 2D Constant False Alarm Rate (CFAR) we can obtain sufficient RA point cloud. Therefore, we adopt the point target tracking scheme based on RA point cloud. One representative of this tracking approach is PmTrack [3], the basic workflow of which is shown in Fig. 7. In each frame, the point cloud is clustered by the DBSCAN algorithm and the centroid of each cluster will be obtained. Then, we compute the euclidean distances between existing trajectories and centroids, and input them as the cost matrix into the Hungarian algorithm to obtain the optimized trajectory-centroid association for trajectory updating. Moreover, the Kalman filtering of constant velocity mode is introduced for trajectory prediction and correction, thereby updating trajectories when they fail to associate and obtaining more accurate results. The trajectory states are categorized into four types. The initial state is candidate, which is converted to active after successive successful associations. After that, the state transitions between active and missing depending on the association result. Trajectories with continuous association

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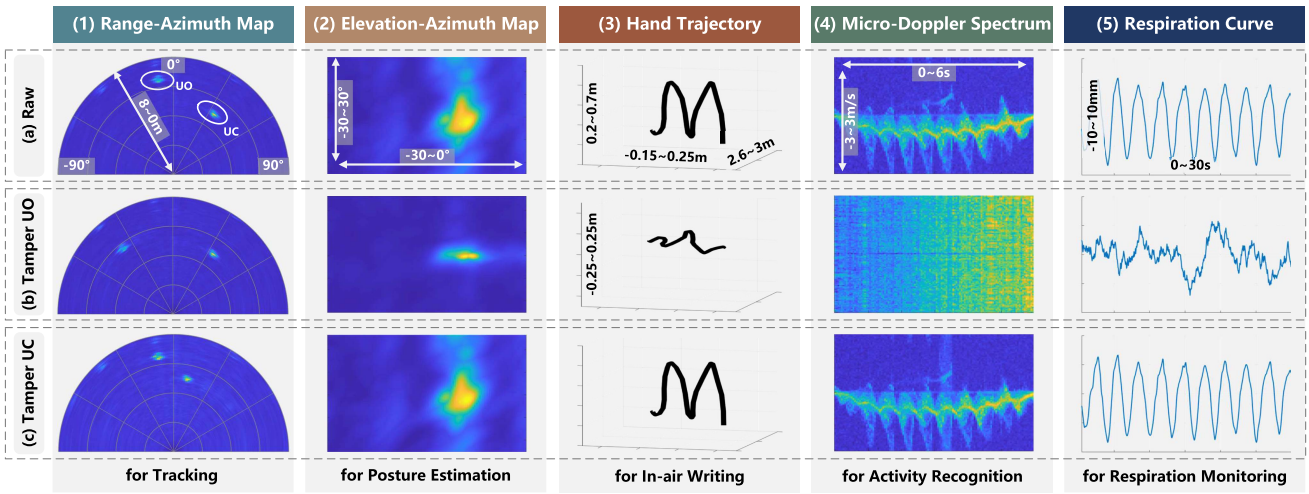


Fig. 8. Demonstration of behavioral privacy filtering in multi-user scenarios. The observation target is always the UO.

failure will be marked as lost. Through this state management, the accuracy of trajectory numbers and the stability of position updates can be effectively ensured. PmTrack also provides a variety of optimization methods that can effectively overcome conventional tracking challenges, such as using angular consistency to screen out ghost clusters caused by multipath effects.

It should be noted that multi-user personalized services require the support of human identification technology, as only then can radar targets be effectively associated with users and their preferences accurately obtained. A wide variety of mmWave-based identification methods have been proposed based on features such as gait [16], point clouds [17], behavioral patterns [18], or vital signs [19]. MuFilter does not rely on specific identification methods and can work well with existing methods to facilitate personalized and secure mmWave sensing. Certainly, to identify individuals based on signal features, it may need to retain relevant behavioral information within the privacy filtering system. To maintain flexibility in privacy protection, user identification and maintenance can be achieved by utilizing additional sensors [3] or employing re-identification techniques [20], along with continuous position data obtained from the tracking module.

B. Subspace Tampering

Next, we take advantage of the radar's spatial separation capabilities to isolate each user's data tampering. The spatial separation capability is reflected in two aspects. First, based on the frequency of the IF signal and the phase vector of the MIMO array, the space can be separated into RA bins by Range FFT and Angle FFT. Second, the energy leakage of the FFT result "spectrum" is limited and acceptable. In other words, the energy of the human reflected signal is concentrated in the RA bins where they are located. Thus, we can isolate each user's signal through RA domain signal extraction, which includes three steps. (1) Perform Range FFT and Angle FFT to obtain RA bins. Note that to ensure the signal can be reversed back to the original signal format, the number of FFT points is the ADC sample number and the array element number, respectively. To

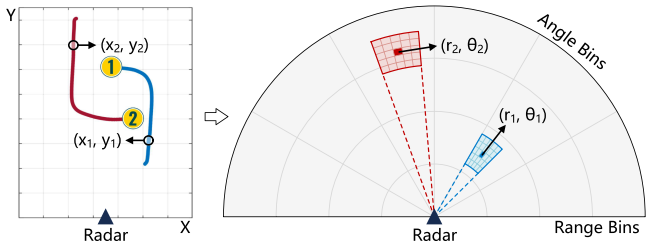


Fig. 9. Position mapping and subspace extraction.

maintain the fidelity and stability of the reversed signal, no windowing or static clutter filtering is performed. (2) Generate subspace through RA bin focus. Take user 1 in Fig. 9 as an example. First, calculate its distance r_1 and azimuth θ_1 relative to the radar based on its position (x_1, y_1). Then, find the RA bin where (r_1, θ_1) is located, and use it as the center bin to spread a reasonable number of bins to serve as the subspace for the user. (3) Perform trampering in each user's subspace. The tampered subspaces will return and form the complete RA domain, and be inverted back to the raw signal format through Angle IFFT and Range IFFT.

In Fig. 8, we validate the effectiveness of subspace tampering in a two-user scenario. For the efficiency of the experiment and the need to control a single variable, we define two types of users: Observation User (UO) and Coexisting User (UC). In each column, we perform the tampering method corresponding to the application on one user, but we only show the sensing signatures of UO. In this way, we can validate the effectiveness of privacy filtering in multi-user scenarios and whether tampering with one user will affect others. In rows (b) and (c), the tampered users are UO and UC, respectively. These two rows respectively verify the effectiveness and isolation of data tampering. As we can see, when UO is tampered with, their signatures are disrupted. When UC is tampered with, UO's signatures remain unaffected. These results indicate that subspace tampering achieves spatial isolation in signal tampering, effectively supporting customizable behavioral privacy protection in multi-user scenarios.

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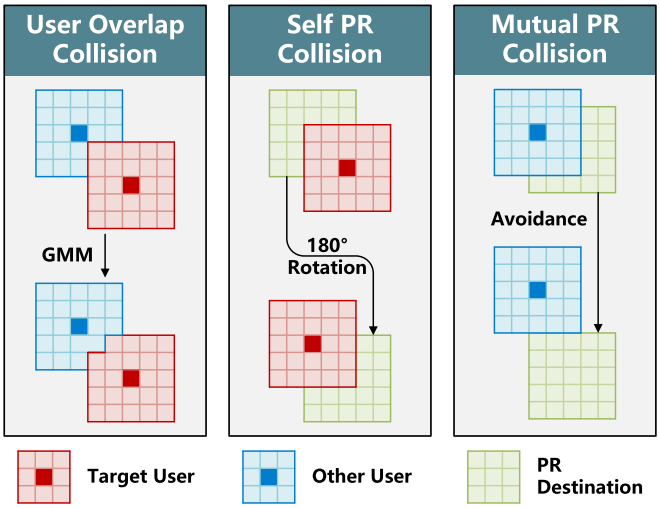


Fig. 10. Subspace collisions and anti-collision solutions.

C. Anti-Collision Mechanism

In the deployment of MuFilter, position collisions may occur in the subspaces of multiple users, which can be categorized into the three types shown in Fig. 10. (1) User overlap collision. When multiple users are close, their subspaces may overlap. We use the Gaussian Mixture Model (GMM) algorithm to cluster the union of RA bins in multiple subspaces to obtain the boundary between subspaces and maintain their isolation. (2) Self-collision of position reset. The PR methods that modify position information can also cause subspace collision. For the user itself, the original subspace and destination subspace may overlap, making it infeasible to exchange the two subspaces to complete PR. In this case, MuFilter fills the non-overlapping bins of the original space with the non-overlapping bins of the destination subspace. This process should minimize the abruptness caused by the filling, thus we adopt the rotation method shown in Fig. 10 to avoid RA bin confusion during the exchange process. (3) Mutual-collision of position reset. PR's destination subspace may also overlap with other users' subspaces. At this moment, we must change the destination to avoid the collision. Considering that there are usually far more range bins than azimuth bins, we suggest sliding the subspace in the range dimension to determine new destinations. In practice, it is better to set a less-passed position as the destination for PR.

In addition to position-related collisions, we observe that MuFilter's local tampering can have a slight impact on high-resolution DoA estimation algorithms such as Conventional Beamforming and Capon. This is because tampering with only part of the azimuth bins affects the covariance matrix of the array signal to some extent. In contrast, if the tampered azimuth bins cover the full dimension, the signal maintains integrity and continuity in the azimuth dimension, and thus does not change the covariance matrix. Therefore, we suggest extending azimuth bins to the entire azimuth dimension when the subspace does not overlap in range with other subspaces.

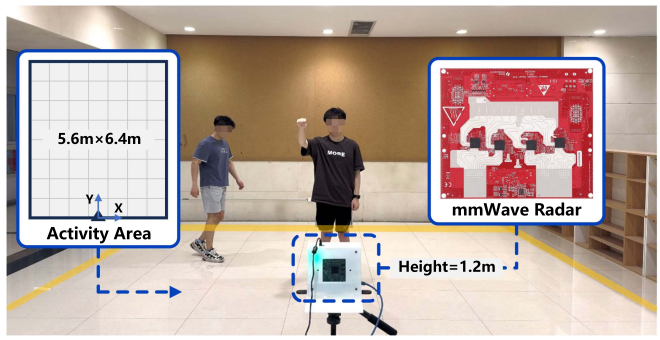


Fig. 11. Experimental environment.

VI. Evaluation

In this section, we conduct extensive experiments to verify the performance of MuFilter. Each participant is fully informed about the experimental content and purpose.

A. Setup

1) Environment: The experimental site is shown in Fig. 11. Our experiment is conducted in a rectangular area of 5.6 m × 6.4 m and the height of radar is 1.2 m.

2) Hardware & Software: We verify the software prototype of MuFilter using a commercial mmWave radar. We use TI MMWCAS-RF-EVM [21] to collect mmWave data. The radar is configured to a 2-chip (#1 and #4) cascade mode, which contains 6 Tx antennas and 8 Rx antennas. The specific parameter settings of the mmWave radar are as follows: the start frequency is set to 77 GHz with an effective bandwidth of 3.6 GHz. The sampling rate is 7.04 MHz, and the frequency slope is 99 MHz/us. Each Tx emits 128 chirps per frame, each chirp contains 256 sampling points, and the frame period is 50 ms. MuFilter and various sensing applications are conducted on MATLAB 2022b.

3) Evaluation Strategy: As before, both UO and UC undergo various tampering modes, and we only obtain sensing signatures of UO and calculate the corresponding metrics. Most experiments are conducted in scenarios involving two users, with one UO and one UC. We design three progressive experimental schemes to evaluate MuFilter in depth from different perspectives.

Single Tampering: This scheme focuses on the effectiveness of each tampering method and whether it impairs non-target sensing applications. Therefore, we disable only one sensing application for one user.

Combined Tampering: Perform multiple tampering methods on only one user. This scheme aims to explore the compatibility between tampering methods and observe whether their simultaneous usage would lead to conflicts. We consider the most extreme scenario, enabling as many tampering methods as possible. As a result, only one sensing application is retained, while all other sensing applications are disabled.

Conflicted Tampering: This scheme validates the core contribution of MuFilter, i.e., when multiple users set different privacy filter modes, whether the enabled sensing applications of each user remain unaffected. Similarly, we consider the most extreme

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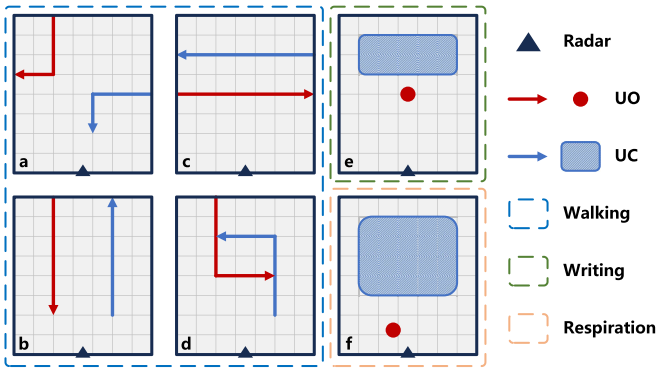


Fig. 12. Preset paths.

scenario. We enable one sensing application for one user while disabling all other sensing applications. The enabled or disabled sensing applications will be disabled or enabled for the other user. In this way, we stimulate the conflicts caused by user variability to the greatest extent.

Additionally, we compare MuFilter with mmFilter and conduct experiments in scenarios involving more users to further validate MuFilter's ability to address the personalized needs of multiple users.

4) Data Collection: Some sensing applications are limited by scenarios. For example, in respiration monitoring, the user's movement will obscure the chest displacement caused by breathing, leading to difficulties in extracting the respiration curve. Dynamic reflections from nearby users can also interfere with the respiration waveform. Therefore, although the tampering methods of MuFilter won't be affected by the deployment scenario, in order to make each sensing result effective and stable, we set three experimental scenarios: walking, in-air writing, and respiration monitoring, as shown in Fig. 12. In each schematic diagram, the red line and point represent the walking trajectory and position of UO, while the blue line and area represent the walking trajectory and free movement area of UC.

Walking: The two users walk along the preset paths. We consider four situations of RA bins overlap: (a) no overlap; (b) overlap in range bins; (c) overlap in angle bins; (d) overlap in both range and angle bins. Under each path setting, we collect data of 6 s for 10 times.

In-air Writing: The UO stands at a distance of 3.2 m from the radar and performs in-air writing. The UC walks freely in the area behind the UO. We collect two hand-drawn patterns, each collecting 6 s for 5 times.

Respiration Monitoring: The UO sits in a chair at (0.96 m, 30°), facing the radar and breathing steadily. The UC walks freely in the area behind the UO. Ten sets of data are collected, each lasting 30 s.

5) Evaluation Metrics: We need quantitative results to evaluate the performance of our system. Although we intuitively demonstrate the impact of data tampering through five sensing signatures in Sections IV and V, these signatures are not necessarily suitable as evaluation objects. Thus, we select the most representative numerical results for each sensing application and set tampering metrics based on them.

TABLE I
CLASSIFICATION OF INTERFERENCE LEVELS

	$E_t\,(m)$	$CR_e\,(\%)$	$S_{ges}\,(\%)$	$SSIM$	$E_{rc}\,(s)$
Unchanged	0-0.08	0-10	95-100	0.95-1	0-0.1
Slightly Interfered	0.08-0.8	10-30	80-95	0.8-0.95	0.1-0.5
Greatly Interfered	0.8-1.5	30-60	50-80	0.5-0.8	0.5-2
Disabled	>1.5	60-100	0-50	0-0.5	>2

Tracking: We implement a standard multi-user tracking algorithm to obtain the moving trajectory of each user. The tamper metric is the euclidean distance of the trajectory coordinates Before Tampering (BT) and After Tampering (AT), i.e., $E_t = \|\mathbf{p}_a - \mathbf{p}_b\|_2$, where $\mathbf{p}_b$ and $\mathbf{p}_a$ represent the coordinates of BT and AT respectively.

Posture Estimation: VIE technique can eliminate information in the vertical dimension. However, the AI model will always output predictions that resemble a normal human body. In other words, posture estimation results such as skeletal coordinates may not accurately reflect the interference effect of MuFilter. Therefore, we obtain the user's 3D point clouds through 2-pass Capon and calculate the standard deviation of their elevation. The tamper metric is the change rate of standard deviation, i.e., $CR_e = |e^a_{std} - e^b_{std}|/e^b_{std}$, where e^b_{std} and e^a_{std} represent the elevation's standard deviation of BT and AT respectively.

In-air Writing: The 3D hand trajectory is projected onto a virtual plane to obtain a 2D image that plots the written character. Then, we use a siamese network to compute the similarity of images of BT and AT, denoted as S_{ges}, which is the tamper metric of this sensing application.

Activity Recognition: For the same considerations as posture estimation, we utilize the sensing signature of activity recognition to evaluate the interference effect. We obtain the user's micro-Doppler spectrum and calculate the Structural Similarity (SSIM) index between the spectrum of BT and AT as the tamper metric. SSIM can better demonstrate changes in image structure, and clearly reflect the interference of MuFilter on velocity information.

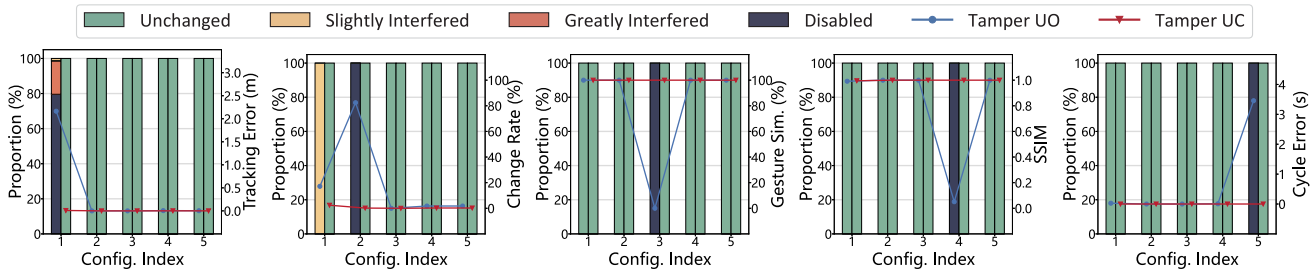
Respiration Monitoring: We focus on the measurement results of respiration rate. Specifically, we first find the peaks of the respiration curve. Then we divide the time difference between the first peak and the last peak Δt by the number of peaks N to obtain the average respiration cycle $T = \Delta t/N$. The tamper metric is the difference between the average respiration cycles of BT and AT, i.e., $E_{rc} = |T_a - T_b|$.

To evaluate the degree of interference on sensing applications, we define four levels of interference: (1) Unchanged. Sensing results are (almost) unchanged. (2) Slightly interfered. Sensing results are changed but still functional. (3) Greatly interfered. Sensing results are severely disrupted that they are not functional. (4) Disabled: Sensing results are completely destroyed. The result metrics and interference level determinations for each sensing application are detailed in Table I.

B. Single Tampering

In this evaluation, we interfere with only one sensing application of one user while leaving the other user undisturbed. To

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(a) Tracking (b) Posture estimation (c) In-air writing (d) Activity recognition (e) Respiration monitoring

Fig. 13. Evaluation results under single tampering.

TABLE II
SINGLE TAMPERING CONFIGURATIONS

Config. Index	Tampering Method			
	PR	VIE	CD	IFPF
1	✓			
2		A/B		
3		×/A		
4			✓	
5				✓

TABLE III
COMBINED TAMPERING CONFIGURATIONS

Config. Index	Tampering Method			
	PR	VIE	CD	IFPF
6		✓	✓	✓
7	✓	×/A	✓	✓
8	✓	A/B	✓	✓
9	✓	✓		✓
10	✓	✓	✓	

align with privacy protection principles, VIE-B which preserves hand information, eliminates the spatial information of the torso required for posture estimation. As mentioned earlier, a user who does not want to reveal their hand trajectory will not want to reveal their posture information either. To independently evaluate posture estimation and in-air writing to meet the preset evaluation strategy, we adopt two VIE modes. The specific configurations are shown in Table II. For the VIE column, we set different VIE modes for posture estimation and in-air writing respectively, where A represents VIE-A, B represents VIE-B, and × represents no VIE. For example, we perform VIE-A and VIE-B in posture estimation and in-air writing respectively in configuration 2. We also follow similar experimental considerations in the subsequent evaluations. The UO and UC serve as the tampered user in turn. We record UO's tamper metrics in Fig. 13. In each subfigure, the bars represent the proportion of different interference levels during the sensing process, while the lines depict specific tamper metrics. In each grouped bar, the left bar and blue line represent the results when UO serves as the tampered user, whereas the right bar and red line represent the results when UC serves as the tampered user. We observe that: (1) Each tampering method of MuFilter can effectively protect user's behavioral privacy. They interfere with the sensing signatures of the tampered user, resulting in the corresponding sensing applications being disabled. For tracking, the proportion of disabled and greatly interfered is greater than 98.6%. The reason why it doesn't achieve 100% disabled is that the UO's actual position is sometimes close to the PR's target position. In other words, PR is actually always effective throughout the user's entire movement process. While for other sensing applications, MuFilter achieves 100% disabled. (2) Each tampering method has excellent isolation, ensuring that they do not affect the normal operation of other sensing applications. In terms of posture estimation, PR introduces slight interference to point cloud generation, causing a change of less than 20% in the elevation's standard deviation. Furthermore, other sensing applications that are not tampered with remain 100% unchanged. (3) MuFilter successfully separates multiple users. When UC serves as the tampered user, all UO's sensing applications are 100% unchanged.

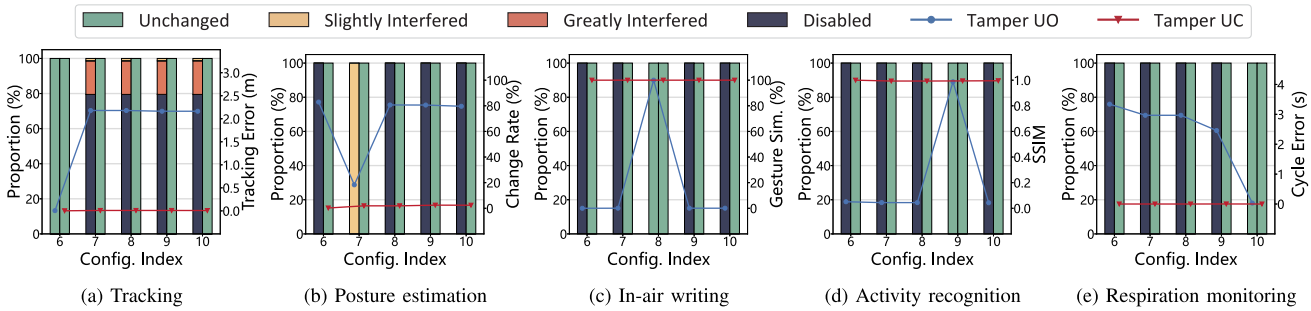
C. Combined Tampering

In practical application scenarios, users may have more diverse privacy protection needs that cannot be adequately addressed by a single tampering method. In such cases, it becomes necessary to combine tampering methods to accommodate diverse privacy protection needs. In this evaluation, we only retain one sensing application and interfere with all other applications of one user while leaving the other user undisturbed. The specific configurations are shown in Table III. The UO and UC serve as the tampered user in turn. The UO's tamper metrics are shown in Fig. 14. As we can see, various tampering methods are compatible with each other, and their combination will not affect the effectiveness of MuFilter. On the one hand, multiple tampering methods can be enabled together to protect multiple behavioral privacy. On the other hand, enabling multiple tampering methods doesn't cause confusion and the retained sensing applications can work normally. In addition, no matter what combination of tampering methods is performed on UC, it will not affect UO.

D. Conflicted Tampering

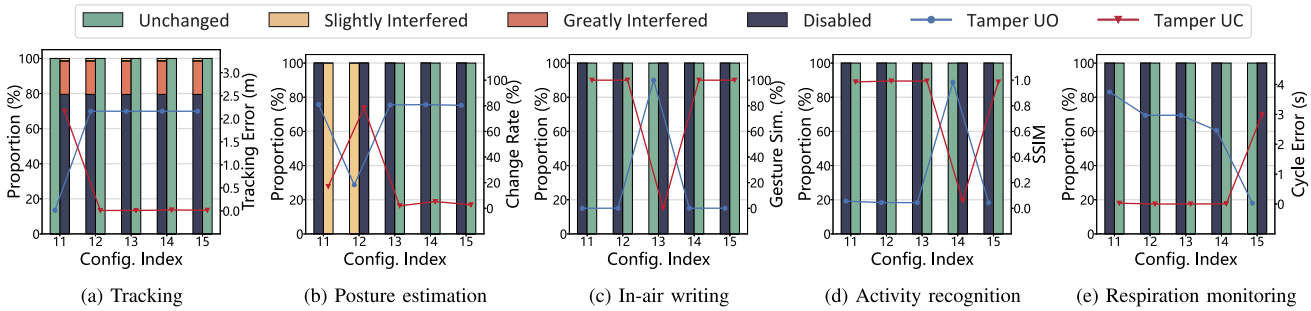
Next, we further evaluate MuFilter's anti-conflict capability in multi-user tasks. We conduct signal tampering on both two users and maximize the degree of conflicts. In this evaluation, we only retain one sensing application and interfere with all other applications of one user while performing completely opposite tampering settings for the other user. The specific configurations

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(a) Tracking (b) Posture estimation (c) In-air writing (d) Activity recognition (e) Respiration monitoring

Fig. 14. Evaluation results under combined tampering.



(a) Tracking (b) Posture estimation (c) In-air writing (d) Activity recognition (e) Respiration monitoring

Fig. 15. Evaluation results under conflicted tampering.

TABLE IV
CONFLICTED TAMPERING CONFIGURATIONS

Config. Index	User Index	Tampering Method			
		PR	VIE	CD	IFPF
11	a		✓	✓	✓
	b	✓			
12	a	✓	✓/A	✓	✓
	b		A/✓		
13	a	✓	A/B	✓	✓
	b		✓/A		
14	a	✓	✓		✓
	b			✓	
15	a	✓	✓	✓	
	b				✓

are shown in Table IV. The UO and UC serve as users a and b in turn. The tamper metrics for UO are shown in Fig. 15. In each grouped bars, the left bar and blue line represent the results when UO serves as user a, whereas the right bar and red line represent the results when UO serves as user b. The results show that the interference degree of each sensing application is in line with its configuration expectations. This demonstrates that even if users' application needs are mutually exclusive, MuFilter can still provide precise privacy protection by retaining required applications while disabling sensitive ones.

E. Conflicted Tampering in Complex Environment

To further verify the effectiveness of MuFilter in real-world scenarios, we conduct the same data collection as in Fig. 11 in the environment shown in Fig. 16. In this environment, the

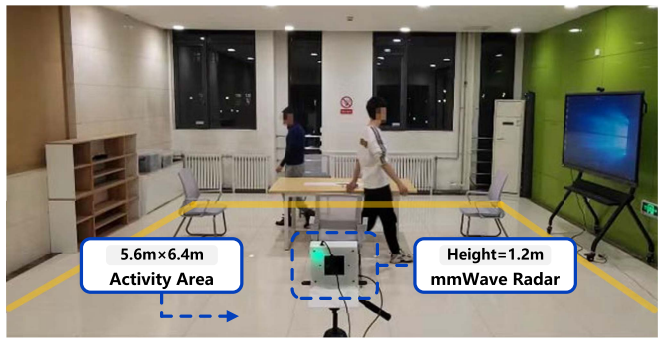


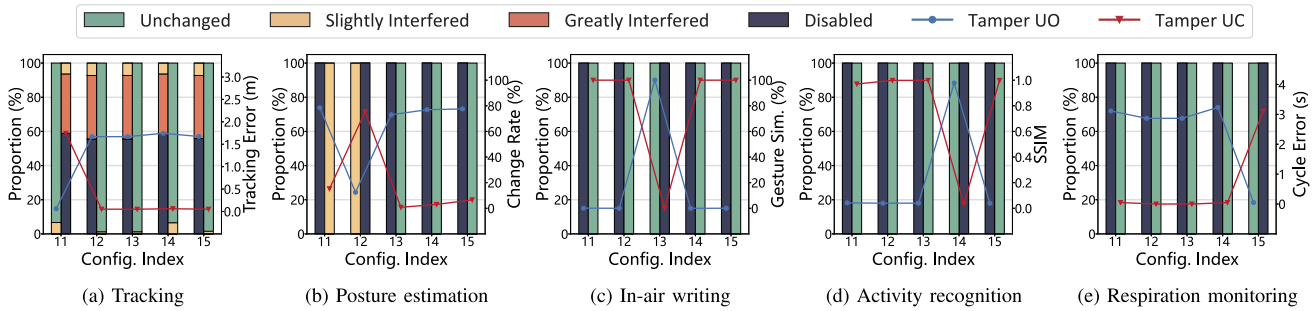
Fig. 16. Complex experimental scenario.

edges of the activity area are adjacent to the walls, and the space is furnished with items such as the table, chair, cabinet, and monitor, effectively simulating the multipath interference in complex real-world environments. We perform conflicted tampering evaluation based on these data, and the results are shown in Fig. 17. Due to the presence of furniture, the usable movement area is reduced, leading to a slight increase in the impact of PR on the non-target user's positions, with an average error of 5.23 cm. This deviation is considered acceptable for human tracking tasks. The evaluation results for other sensing tasks are similar to those obtained in the hall scenario (Fig. 15), demonstrating that MuFilter can effectively operate in real-life and workplace environments.

F. Performance Under More Users

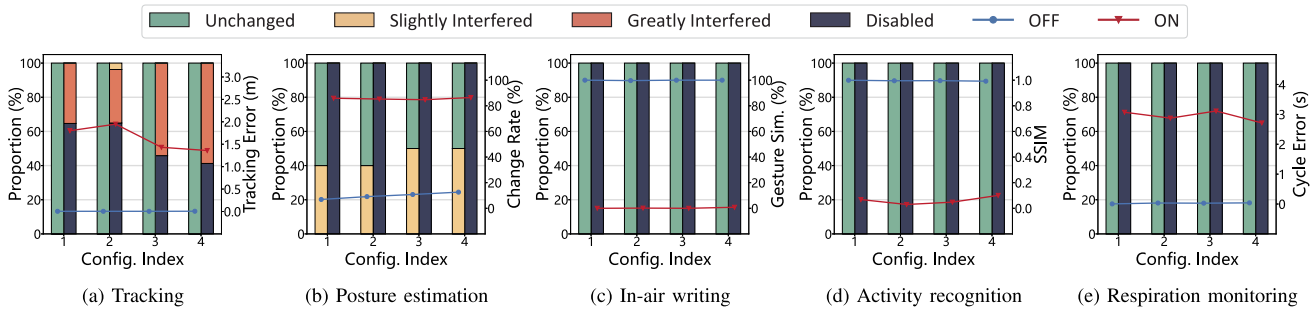
We evaluate the personalized privacy protection performance of MuFilter for more users. The number of users in the scene varies from 1 to 4, with one user selected as UO and the

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(a) Tracking (b) Posture estimation (c) In-air writing (d) Activity recognition (e) Respiration monitoring

Fig. 17. Evaluation results under conflicted tampering in the comlex environment.



(a) Tracking (b) Posture estimation (c) In-air writing (d) Activity recognition (e) Respiration monitoring

Fig. 18. Evaluation results under multi-user scenarios.

remaining users as UC. To fully satisfy personalized requirements, we randomly combine each user's application settings. For each user, the five sensing applications are randomly set to be enabled or disabled with a probability of 50%. In particular, enabling posture estimation triggers in-air writing to be enabled, while disabling in-air writing triggers posture estimation to be disabled. Subsequently, MuFilter automatically combines the corresponding tampering methods. The results are shown in Fig. 18. During the evaluation, we manually set the specific sensing applications of UO to be enabled or disabled. For example, in each bar group in Fig. 18(a) which represents tracking, the tracking of UO is enabled and disabled in the left bar and right bar respectively. We find that for tracking, in-air writing, activity recognition and respiration monitoring, the increase in the number of users essentially does not affect the experimental results. In Fig. 18(b), as the number of users increases, the change rate of the standard deviation of the point cloud's elevation gradually increases, growing from 6.89% to 12.61%. This is because when there are more users in the space, the overlap between them intensifies, leading to spatial information distortion during signal tampering. However, this distortion remains slight and does not significantly affect posture estimation. The above results demonstrate that our system achieves personalized privacy protection in multi-user scenarios successfully. Additionally, in the indoor environment examined in the experiment, the tracking system can accurately track four moving targets, which is sufficient for typical household scenarios. However, as the number of individuals increases, the probability of obstruction, overlap, and multipath interference also escalates, which may result in trajectory errors. We recommend placing the radar at the upper corner of the wall to reduce the impact of obstructions and multipath propagation, and adopt flexible point/group fusion target detection schemes to cope with frequent point cloud overlaps, thus maintaining reliable multi-user tracking performance.

TABLE V
METRIC COMPARISON WITH MMFILTER

Metric	E_t (m)	CR_e (%)	$SSIM$	E_{rc} (s)
mmFilter	0.298	44.71	0.0487	4.10
MuFilter	0.009	3.85	0.9999	0.00

G. Comparison With mmFilter

We compare the performance of our system with mmFilter. mmFilter uses the idea of signal reversion to perform application-oriented privacy filtering on the signal. It only considers single-user scenarios, thus the signal tampering will have a global impact on the radar signal. In this comparison, we omit in-air writing because mmFilter's hand-preserving mechanism enables it to provide the same results as MuFilter when the range bins of multiple users do not overlap. Therefore, our comparative experiments involve four sensing applications: tracking, activity recognition, posture estimation, and respiration monitoring, corresponding to four metrics. We disable each application for UC and observe the corresponding metric for UO. For example, for tracking, we perform PR on UC and observe the tracking error of UO. The experimental results are shown in Table V. Since mmFilter does not consider the presence of multiple users in the scene, its performance is unsatisfactory across the four evaluation metrics. In other words, when we perform signal tampering to any user, it will affect the normal use of sensing

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TABLE VI
TRACKING ERROR UNDER DIFFERENT PRESET PATHS

Preset Path	a (m)	b (m)	c (m)	d (m)
mmFilter	0.067	0.392	0.155	0.577
MuFilter	0.002	0.014	0.005	0.013

TABLE VII
TIME COST COMPARISON WITH MMFILTER

Method	PR (ms)	VIE (ms)	CD (ms)	IFPF (ms)
mmFilter	39.33	12.44	6.34	3.21
MuFilter	15.76	7.25	4.58	0.73
Improvement	59.93%	41.72%	27.76%	77.26%

applications for other users. In contrast, MuFilter isolates the reflection signals for each user, thereby ensuring its effectiveness in multi-user scenarios.

When performing VIE, CD, and IFPF, mmFilter's global processing mechanism will inevitably affect other users. Furthermore, mmFilter's PR also causes a deviation of 0.298 m in UO's position, as the PR of mmFilter leads to compression and displacement of other range and angle bins in space. The tracking error in the four preset walking paths are shown in Table VI. We find that under path a, where there is no overlap in range and angle bins, the tracking error is only 0.067 m. However, under path d, where there is overlap in both range and angle bins for both users, the average tracking error increases to 0.577 m. In MuFilter, we design a new PR mechanism that can accurately separate the signals of two users in space, and will not affect other parts of the space when the position is moved. As a result, under path d, MuFilter can still achieve a tracking error of 0.013 m.

In addition to the advantage of supporting multi-user scenarios, MuFilter's subspace tampering strategy reduces the region of tampered data, thereby reducing computation time. The time cost of performing one tampering with mmFilter and MuFilter, using an Intel i7-11700 2.50 GHz CPU, is recorded in Table VII,

which demonstrates that the subspace tampering strategy can reduce the computation time by 27% to 77%, thus enabling more efficient service for multiple users.

VII. RELATED WORK

A. RF-Based Sensing Security and Privacy

The application of radio frequency (RF) technology is becoming increasingly widespread, bringing along with it concerns regarding security and privacy. By measuring the subtle vibrations of objects, mmWave [22], [23], [24] and RFID [25] technologies can eavesdrop on human speech. Information leakage from smartphones and screens poses a significant threat to sensitive personal information such as passwords [26], [27], [28]. Additionally, leakage of user location information is also a serious problem. Many studies use through-wall radar to obtain users' location [29], [30], leading to the risk of burglary. Some physical layer attack methods [10], [11], [31] perform appearing-disappearing spoofing through special physical devices, thereby tampering with the target's existence or location information. RF-Protect [2] deploys reflectors in the environment to inject dummy signals into the backscattered signal to deceive potential attackers. Radar^2 [32] allows users to utilize a single mmWave radar to detect the presence of other mmWave radars and obtain their locations. However, these works can't achieve customizable tampering on signals to meet the user's different behavioral privacy needs. mmFilter [12] can protect the behavioral privacy of single user through customizable tampering on mmWave signals. However, the signal tampering scheme of mmFilter is global and thus fails multi-user scenarios. Our work performs subspace tampering on mmWave signals and considers the conflicts in multi-user scenarios, allowing multiple users to customize behavioral privacy protection according to their own needs.

B. mmWave-Based Human Sensing

The continuous development of mmWave technology supports an increasing number of applications in human sensing. The high spatial resolution enables precise multi-person tracking [3], [33], human imaging [34], skeleton estimation [4], [35], and mesh reconstruction [5], [36]. Human activity recognition such as gesture recognition [7], [37], [38] and in-air hand tracking [6], [39], [40] is a promising technology in human-computer interaction. Gait [16], point clouds [17], behavioral patterns [18], vital signs [19], facial contours [41], voiceprints [42], or with the help of additional sensors [3], [20], user identity recognition can be achieved. It involves activity classification using micro-Doppler [8] or point cloud information [43]. Utilizing the sensitivity to small displacements, mmWave radar can empower health applications such as breathing and heartbeat monitoring [9], [44], [45]. In recent years, some works propose signal augmentation methods based on signal simulation to reduce the burden of collecting training data [13], [46], [47]. The widespread use and powerful sensing capabilities of mmWave make it possible to extract user behavioral privacy by stealing and analyzing raw signals. Our work proposes signal tampering schemes applicable to various privacy protection requirements, effectively addressing privacy concerns caused by the leakage of radar data.

VIII. CONCLUSION

This paper studied the behavioral privacy leakage threat of mmWave radar sensing and presented MuFilter, a data masking system for multi-user customizable behavioral privacy protection. MuFilter adopts the idea of dimensional signal interference and performs targeted signal tampering based on users' sensing service needs and privacy protection needs. The introduction of multi-person tracking technology and the proposal of subspace tampering technique allow MuFilter to isolate each user's signal and prevent data tampering for one user from affecting other users. Extensive experiments evaluated the performance of MuFilter under various multi-user personalized privacy protection settings, which shows that it can interfere with targeted behavioral signatures with a 100% success rate, while the degree of impact on other users' signatures ranges from 0% to 3.85%.

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REFERENCES

[1] J. Zhang et al., "A survey of mmWave-based human sensing: Technology, platforms and applications," *IEEE Commun. Surv. Tut.*, vol. 25, no. 4, pp. 2052–2087, Fourth Quarter 2023.

[2] J. Shenoy, Z. Liu, B. Tao, Z. Kabelac, and D. Vasisht, "RF-Protect: Privacy against device-free human tracking," in *Proc. ACM SIGCOMM Conf.*, 2022, pp. 588–600.

[3] H. Liu, X. Liu, X. Xie, X. Tong, and K. Li, "PmTrack: Enabling personalized mmWave-based human tracking," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 4, pp. 1–30, 2023.

[4] H. Kong et al., "m3Track: mmWave-based multi-user 3D posture tracking," in *Proc. ACM Annu. Int. Conf. Mobile Syst. Appl. Serv.*, 2022, pp. 491–503.

[5] H. Xue et al., "M4esh: mmWave-based 3D human mesh construction for multiple subjects," in *Proc. ACM Conf. Embedded Netw. Sensor Syst.*, 2022, pp. 391–406.

[6] C. Lin et al., "AirWrite: An aerial handwriting trajectory tracking and recognition system with mmWave," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 13 325–13 341, Dec. 2024.

[7] Y. Li et al., "Towards domain-independent and real-time gesture recognition using mmWave signal," *IEEE Trans. Mobile Comput.*, vol. 22, no. 12, pp. 7355–7369, Dec. 2023.

[8] S. Zhu, R. G. Guendel, A. Yarovoy, and F. Fioranelli, "Continuous human activity recognition with distributed radar sensor networks and CNN-RNN architectures," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5115215.

[9] G. Paterniani et al., "Radar-based monitoring of vital signs: A tutorial overview," in *Proc. IEEE*, vol. 111, no. 3, pp. 277–317, Mar. 2023.

[10] X. Chen et al., "MetaWave: Attacking mmWave sensing with meta-material-enhanced tags," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2023, pp. 1–17.

[11] R. Reddy Vennam et al., "mmSpoof: Resilient spoofing of automotive millimeter-wave radars using reflect array," in *Proc. IEEE Symp. Secur. Privacy*, 2023, pp. 1807–1821.

[12] H. Liu, X. Liu, X. Xie, X. Tong, T. Shi, and K. Li, "Application-oriented privacy filter for mmWave radar," *IEEE Commun. Mag.*, vol. 61, no. 12, pp. 168–174, Dec. 2023.

[13] X. Zhang, Z. Li, and J. Zhang, "Synthesized millimeter-waves for human motion sensing," in *Proc. ACM Conf. Embedded Netw. Sensor Syst.*, 2022, pp. 377–390.

[14] M. U. Hassan, M. H. Rehmani, and J. Chen, "Differential privacy techniques for cyber physical systems: A survey," *IEEE Commun. Surv. Tut.*, vol. 22, no. 1, pp. 746–789, First Quarter 2020.

[15] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," 2014.

[16] J. Pegoraro, F. Meneghello, and M. Rossi, "Multiperson continuous tracking and identification from mm-Wave micro-doppler signatures," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 2994–3009, Apr. 2021.

[17] Y. Cheng and Y. Liu, "Person reidentification based on automotive radar point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5101913.

[18] C.-Y. Hsu, R. Hristov, G.-H. Lee, M. Zhao, and D. Katabi, "Enabling identification and behavioral sensing in homes using radio reflections," in *Proc. ACM Annu. ACM Conf. Hum. Factors Comput. Syst.*, 2019, pp. 1–13.

[19] Y. Wang, T. Gu, T. H. Luan, and Y. Yu, "Your breath doesn't lie: Multi-user authentication by sensing respiration using mmWave radar," in *Proc. 19th Annu. IEEE Int. Conf. Sens. Commun. Netw.*, 2022, pp. 64–72.

[20] D. Cao, R. Liu, H. Li, S. Wang, W. Jiang, and C. X. Lu, "Cross vision-RF gait re-identification with low-cost RGB-D cameras and mmWave radars," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 3, pp. 1–25, 2022.

[21] Texas Instruments, "AWRx cascaded radar RF evaluation module (MMWCAS-RF-EVM)," 2020. [Online]. Available: https://www.ti.com/lit/ug/swru553a/swru553a.pdf

[22] S. Basak and M. Gowda, "mmSpy: Spying phone calls using mmWave radars," in *Proc. IEEE Symp. Secur. Privacy*, 2022, pp. 1211–1228.

[23] C. Wang et al., "mmEve: Eavesdropping on smartphone's earpiece via COTS mmWave device," in *Proc. ACM Annu. Int. Conf. Mobile Comput. Netw.*, 2022, pp. 338–351.

[24] P. Hu, W. Li, R. Spolaor, and X. Cheng, "mmEcho: A mmWave-based acoustic eavesdropping method," in *Proc. IEEE Symp. Secur. Privacy*, 2023, pp. 1840–1856.

[25] C. Wang et al., "Thru-the-wall eavesdropping on loudspeakers via RFID by capturing sub-mm level vibration," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 5, no. 4, pp. 1–25, 2021.

[26] Z. Li et al., "WaveSpy: Remote and through-wall screen attack via mmWave sensing," in *Proc. IEEE Symp. Secur. Privacy*, 2020, pp. 217–232.

[27] Z. Li et al., "SpiralSpy: Exploring a stealthy and practical covert channel to attack air-gapped computing devices via mmWave sensing," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2022, pp. 1–16.

[28] J. Hu et al., "Password-stealing without hacking: Wi-Fi enabled practical keystroke eavesdropping," in *Proc. ACM Conf. Comput. Commun. Secur.*, 2023, pp. 239–252.

[29] G. Gennarelli, G. Vivone, P. Braca, F. Soldovieri, and M. G. Amin, "Multiple extended target tracking for through-wall radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6482–6494, Dec. 2015.

[30] H. Li, G. Cui, L. Kong, G. Chen, M. Wang, and S. Guo, "Robust human targets tracking for MIMO through-wall radar via multi-algorithm fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1154–1164, Apr. 2019.

[31] Z. Sun, S. Balakrishnan, L. Su, A. Bhuyan, P. Wang, and C. Qiao, "Who is in control? Practical physical layer attack and defense for mmWave-based sensing in autonomous vehicles," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 3199–3214, 2021.

[32] Y. Qiu, J. Zhang, Y. Chen, J. Zhang, and B. Ji, "Radar2: Passive spy radar detection and localization using COTS mmWave radar," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2810–2825, 2023.

[33] W. Chen et al., "Environment-aware multi-person tracking in indoor environments with mmWave radars," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 3, pp. 1–29, 2023.

[34] A. Adhikari, H. Regmi, S. Sur, and S. Nelakuditi, "MiShape: Accurate human silhouettes and body joints from commodity millimeter-wave devices," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 3, pp. 1–31, 2022.

[35] S. An and U. Y. Ogras, "MARS: mmWave-based assistive rehabilitation system for smart healthcare," *ACM Trans. Embedded Comput. Syst.*, vol. 20, no. 5s, pp. 1–22, 2021.

[36] H. Ding et al., "MI-Mesh: 3D human mesh construction by fusing image and millimeter wave," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 1, pp. 1–24, 2023.

[37] H. Wei et al., "IndexPen: Two-finger text input with millimeter-wave radar," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 2, pp. 1–39, 2022.

[38] H. Liu et al., "mTransSee: Enabling environment-independent mmWave sensing based gesture recognition via transfer learning," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 1, pp. 1–28, 2022.

[39] W. Li, T. He, N. Jing, and L. Wang, "mmHSV: In-air handwritten signature verification via millimeter-wave radar," *ACM Trans. Internet Things*, vol. 4, pp. 1–22, 2023.

[40] S. D. Regani, C. Wu, B. Wang, M. Wu, and K. J. R. Liu, "mmWrite: Passive handwriting tracking using a single millimeter-wave radio," *IEEE Internet Things J.*, vol. 8, no. 17, pp. 13 291–13 305, Sep. 2021.

[41] W. Xu et al., "Mask does not matter: Anti-spoofing face authentication using mmWave without on-site registration," in *Proc. ACM Annu. Int. Conf. Mobile Comput. Netw.*, 2022, pp. 310–323.

[42] H. Li et al., "VocalPrint: Exploring a resilient and secure voice authentication via mmWave biometric interrogation," in *Proc. ACM Conf. Embedded Netw. Sensor Syst.*, 2020, pp. 312–325.

[43] D. Zhang et al., "LT-Fall: The design and implementation of a life-threatening fall detection and alarming system," in *Proc. ACM Interactive, Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 1, pp. 1–24, 2023.

[44] X. Xu et al., "mmECG: Monitoring human cardiac cycle in driving environments leveraging millimeter wave," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2022, pp. 90–99.

[45] Y. Wang, Z. Wang, J. A. Zhang, H. Zhang, and M. Xu, "Vital sign monitoring in dynamic environment via mmWave radar and camera fusion," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4163–4180, May 2024.

[46] K. Deng, D. Zhao, Z. Zhang, S. Wang, W. Zheng, and H. Ma, "Midas++: Generating training data of mmWave radars from videos for privacy-preserving human sensing with mobility," *IEEE Trans. Mobile Comput.*, vol. 23, no. 6, pp. 6650–6666, Jun. 2024.

[47] K. Ahuja, Y. Jiang, M. Goel, and C. Harrison, "Vid2Doppler: Synthesizing Doppler radar data from videos for training privacy-preserving activity recognition," in *Proc. ACM Annu. ACM Conf. Hum. Factors Comput. Syst.*, 2021, pp. 1–10.

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